

PROJECT DOCUMENTATION

Ridelytics

Smart Transportation Business Intelligence System

Project Title	Ridelytics - Smart Transportation Business Intelligence System
Domain	Smart Transportation / Ride-Hailing
Primary Tool	Power BI Desktop & Service, SQL Server, SSIS, SSAS, SSRS, n8n
Context	Information Technology Institute (ITI) – Graduation Project
Academic Year	2025 / 2026
Target Market	Egyptian Ride-Hailing Operators
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1. Project Title

Full Title	Ridelytics – Smart Transportation Business Intelligence & Analytics Platform
Subtitle	End-to-End BI System for Egyptian Ride-Hailing Operators

The name Ridelytics was chosen to reflect the fusion of ride-hailing operations and intelligent, data-driven decision-making. The platform is purpose-built for operators competing in Egypt's dynamic transportation market alongside global players such as Uber, InDrive, Careem, DiDi, and Bolt.

2. Business Understanding

2.1 Market Context

Egypt's ride-hailing market has experienced rapid growth over the past decade, driven by urbanization, smartphone penetration exceeding 60%, and a young population of approximately 105 million. Cairo alone accounts for over 20 million residents, creating one of the highest urban-mobility demand densities in the MENA region. According to industry estimates, the Egyptian ride-hailing market is projected to surpass USD 1.2 billion in gross merchandise value by 2026, with compound annual growth in the double digits.

Operators face intense competitive pressure from global platforms (Uber, Careem, InDrive, DiDi, Bolt), each offering aggressive promotional pricing and localized features such as Arabic-language interfaces, cash payment acceptance, and Ramadan-specific campaigns. Differentiation increasingly depends not on technology alone, but on the speed and quality of data-driven decisions made by operations, finance, and executive teams.

2.2 System Description

The Ridelytics Analytics Platform is a comprehensive, end-to-end Business Intelligence solution designed for every ride-hailing operator providing the insights and analytics needed to understand, optimize, and grow their operations. Built to serve the full operational scope of a ride-hailing business, the platform consolidates data from multiple operational sources trip records, driver profiles, passenger accounts, payment gateways, GPS telemetry, and customer-support tickets into a unified data warehouse. From this warehouse, 6 SSRS reports, 20+ dashboards, Power BI, and AI-powered automation workflows are delivered to stakeholders across the organization, giving every operator a clear, real-time view of their system.

2.3 Core Business Problems Addressed

The platform is designed to solve the following critical business challenges:

- Fragmented data across trip, driver, payment, and support systems with no unified view.
- Inability to identify high-churn passenger segments before they permanently leave the platform.
- Lack of real-time visibility into driver supply vs. passenger demand gaps by geographic zone.
- Manual complaint triage results in slow escalation of critical safety incidents.
- No systematic comparison of pricing competitiveness against Uber, InDrive, and Bolt in Egyptian zones.
- Insufficient insight into seasonal demand shifts during Ramadan, Eid, school seasons, and summer.

- Limited financial granularity difficulty attributing net margin to specific zones, vehicle types, or time bands.

3. Project Objectives

3.1 Technical Objectives

- Design and implement a normalized transactional OLTP database with full ERD documentation, relational schema mapping, and a detailed data dictionary covering core tables hosted on Azure Cloud SQL Database.
- Build a galaxy schema data warehouse with dimension and fact tables, populated via SSIS ETL pipelines with SCD handling and surrogate key resolution, and analyzed via SSAS multidimensional cubes.
- Develop stored procedures covering all CRUD operations (SELECT, INSERT, UPDATE, DELETE) across all core OLTP tables, with built-in validation logic.
- Implement an SSRS operational reporting layer delivering 6 structured reports
- Publish 20+ Power BI dashboards across 5 functional domains to Power BI Service, accessible through a navigation hub with a consistent Figma-designed theme.
- Integrate 4 AI automation pipelines using n8n for intelligent workflow orchestration.
- Build and deploy the Ridelytics web platform, featuring a public landing page, a secure JWT-authenticated admin portal with role-based access, full CRUD management for all operational entities, a live executive dashboard with KPIs and charts, and an embedded AI chat assistant (Cognitive Core) integrated via n8n webhook supporting Arabic and English natural language queries.

3.2 Strategic Objectives

- Build a scalable, production-grade BI ecosystem that unifies all ride-hailing operational data into a single source of truth.
- Deliver actionable, role-specific insights to every level of the organization from board-level executive summaries to zone-level operational heat maps.
- Enable proactive decision-making by providing trend analysis, predictive retention scoring, and anomaly detection rather than purely historical reporting.
- Reduce executive reporting time from manual cycles to automated AI-narrated summaries delivered via email.

3.3 Analytical Objectives

- Trend Analysis: Identify revenue, trip volume, and driver utilization trends across daily, weekly, monthly, and seasonal dimensions.
- Demand Forecasting: Surface zone-level demand patterns to support driver incentive programs and pre-positioning strategies.
- Rider Lifecycle Analytics: Track rider cohorts from first trip through retention, dormancy, and re-engagement.
- Financial Performance: Decompose gross revenue into net margin after driver payouts, promotions, and operational costs.
- Geographic Intelligence: Visualize Egypt's 27 governorates by revenue density, complaint rate, and unfulfilled demand.

3.4 Operational Objectives

- **Decision Support:** Reduce the time for operations managers to identify supply shortfalls from hours to minutes.
- **Complaint Management:** Automate complaint classification and escalation to eliminate manual triage bottlenecks and streamline processes.
- **Driver Retention:** Provide visibility into driver churn signals and the effectiveness of retention incentives.
- **Rider Retention:** Deliver actionable insights into early churn indicators and measure the performance of promotional incentives.
- **AI Automation:** Deploy four AI-powered workflows to automate passenger re-engagement, complaint classification, BI narration, and natural language querying.

4. Reports

The platform delivers 6 structured reports via SQL Server Reporting Services (SSRS). Each report targets a specific operational or financial need and is designed for scheduled distribution to the relevant stakeholder group.

1. Operations Summary:

A snapshot covering total trip count, completed vs. cancelled trips, average trip duration, zone-level demand, driver online hours, and peak-hour analysis. Includes exception flags for cancellation rate spikes or zones with zero driver availability.

2. Driver Performance Report:

A report evaluating each driver across KPIs: trips completed, average passenger rating, cancellation rate, and earnings per hour.

3. Riders List Report:

A basic customer overview report displaying rider activity, including total trips taken, total spending, and the most recent trip date. Helps in understanding user engagement levels and identifying active versus inactive riders for retention efforts.

4. Financial Report:

A clean financial snapshot showing total revenue generated , along with discounts applied and final net revenue. Provides a simple breakdown for tracking earnings and monitoring the impact of discounts on overall revenue.

5. Trips Details Report:

A detailed transactional report listing individual trips with key attributes such as trip ID, driver, passenger, distance, fare, and trip status. Enables granular analysis and is commonly used for auditing, troubleshooting, and operational review.

6. Promotions Report:

A post-promo summary that measures success based on usage, costs, and ROI. It compares promo users to non-promo users to find the true impact of the discount, and highlights which promo types perform best for different passenger segments.

5. Dashboards

The platform delivers 21 Power BI dashboards organized into five functional domains: Executive, Operational, Financial, Customer, and Geographic Intelligence. All dashboard is published to Power BI Service.

1. Executive Overview:

Top-level KPIs for the CEO, like revenue, trips, growth, and market pulse in one view.

2. Operations Overview:

Tracks trip zones, total active drivers, and unfulfilled demand.

3. Revenue & Profit:

Where the money comes from and where it goes fares, driver payouts, fees, and net margin.

4. Driver Performance:

How drivers are doing: trips completed, ratings, earnings, and cancellation rates.

5. Riders Insights:

Who your riders are, how often they ride, how much they spend, and who's about to leave.

6. Demand by Zone:

Which areas are busiest by hour, and where is demand going unfulfilled.

7. Surge Pricing:

When and where the surge kicks in, how high it goes, and how much extra revenue it generates.

8. Promo Performance:

Which discount campaigns are working and which are just burning money.

9. Complaints & Quality:

What passengers are complaining about, how fast issues get resolved, and which zones are problem areas.

10. Driver Supply:

How many drivers are active, how many are leaving, and what incentives are keeping them.

11. Zone Map Analytics:

A visual map of Egypt governorates colored by revenue, demand, or complaints, with drill-down capability for deeper analysis.

12. Vehicles Overview:

Vehicle types, model year, and inspection status.

13. Expansion Opportunities:

Governorates with high demand but low driver supply are where to grow next.

14. Seasonal Patterns:

How demand shifts during Ramadan, holidays, and different times of the year.

15. Payment Methods:

How much is cash vs wallet vs card, and how that split is changing over time.

16. Trip Flow Map:

Which zones do people travel between, mostly the busiest corridors in the city.

17. Driver Location Clusters:

Where drivers live vs where they actually work, spotting supply gaps by area.

18. Fare vs Distance:

Are you charging consistently? Each trip is plotted by fare and distance to spot outliers.

19. Rider Churn & Ratings:

Of riders who joined each month, how many are still riding 3, 6, or 12 months later.

20. Cancellation Tracker:

Cancellation rate by reason, zone, and time identifying who cancels more and where intervention is needed.

21. Pickup Experience:

How quickly drivers reach riders, focusing on pickup speed and waiting time. Helps identify delays and improve efficiency in the pickup process.

6. AI Features

The platform embeds four AI-powered automation workflows orchestrated via n8n (primary) with LangFlow as an alternative. Each workflow targets a distinct operational pain point, combining SQL data extraction, large language model reasoning (Claude / GPT-4o), and automated notification delivery. All workflows operate on a scheduled or event-driven basis and are fully auditable.

Workflow 1 — Rider Churn Re-Engagement

Objective: Automatically identify dormant high-value passengers and re-engage them with a personalized Arabic message and a unique promotional code before they churn permanently.

Workflow 2 — Complaint Auto-Classification & Escalation

Objective: Eliminate manual complaint triage by automatically classifying every inbound complaint, assessing severity, and immediately escalating critical safety incidents to the Compliance Officer.

Workflow 3 — Daily BI Insight Narrator

Objective: Deliver an AI-generated, human-readable executive summary of the day's key performance metrics, eliminating the need for manual report review.

Workflow 4 — Natural Language Query Interface (Arabic & English)

Objective: Enable operations managers to query the data warehouse in plain Arabic or English without writing SQL, democratizing data access across non-technical stakeholders.

6. Website & Admin Portal

The platform includes a fully deployed web application connected to SQL Server and integrated with n8n via webhook. The website serves as the operational command center for the Ridelytics platform, combining a public-facing landing page with a secure, role-based admin portal.

1. Landing Page

Present the Ridelytics platform to stakeholders with a professional product overview, highlighting key system metrics, the five analytical domains, and direct access to the admin portal.

2. Admin Dashboard

Deliver a real-time executive overview of platform health, displaying live KPIs including total trips, revenue, active drivers, average rating, and cancellation rate, alongside a live trip feed and driver leaderboard.

3. Entity Governance (CRUD Management)

Enable administrators to perform full create, read, update, and delete operations across all 12 operational entities including drivers, riders, trips, vehicles, payments, complaints, and more, through a structured, searchable management interface.

4. AI Assistant

Empower non-technical users to query the data warehouse in plain Arabic or English without writing SQL, receiving instant results rendered as charts or data tables directly within the portal.